

Curriculum Vitae

Technology Specialist at the European Commission's EU AI Office, with a background in AI safety research and quantum computing. Research interests include evaluations of autonomous AI systems, goal drift, reward hacking, and robustness of safety mechanisms. Experience mentoring AI safety research through SPAR, Algoverse, and AI Safety Camp.

Professional Experience

May 2026–present **Technology Specialist, EU AI Office, European Commission**, Brussels, Belgium.

Education

- 2020–25 **Physics PhD (Quantum Computing)**, *Instituto Superior Técnico (IST)*, Portugal.
Quantum algorithms for nonlinear PDEs, quantum error correction, and near-term (NISQ) device applications.
Supervisor: Prof. Yasser Omar.
- 2025 **Visiting Scholar**, *UC Berkeley*, USA.
- 2024 **Visiting Student**, *MIT*, USA.
- 2019–20 **Graduate Research Assistant**, *Instituto de Telecomunicações*, Portugal.
- 2014–19 **Integrated Master's Degree (MSc) in Engineering Physics (MEFT)**, *IST*, Lisbon.
Bachelor's and Master's GPA of 18.4 and 18.1/20, resp.; **one of the top students in the course**.
Academic Excellence Diploma (2014/15, 2015/16); Academic Merit Diploma (2016/17, 2018/19); 1st Honorable Mention of "Academic Excellence in MEFT" (2016/17).
- 2018 **Erasmus at École Polytechnique Fédérale de Lausanne (EPFL)**, Switzerland.

AI Safety Research

- 2025 **CHAI Internship**, *UC Berkeley*.
Developed methods to train more robust and performant probes when labeled samples are limited.
- 2025 **UK AISI Bounty Programme**, *Contractor*.
Implemented improved scaffolding to study agents solving complex tasks.
- 2024–25 **Catalyze AI Safety Incubation Program, Phases 1 & 2**, London.
Explored safety evaluations for AI agents, founding Anchor Research (no longer active).
- 2024 **Pivotal Research Fellowship**, *Pivotal*.
Studied learned look-ahead behavior in chess neural networks to understand planning in neural networks.
- 2024 **Research Engineers Club**, *Safe AI London*.
Replicated WMDP benchmark paper.
- 2023 **AI Safety Hub Labs**, *Team Leader*, Oxford.
Investigated how RL fine-tuning biases language models towards more extractable features.

Mentorship & Field-Building

- 2026 **SPAR Spring 2026**, *Mentor*.
2 projects, 8 mentees: (1) applying Fluid Benchmarking to efficiently run AI safety benchmarks; (2) operationalizing situational awareness in LLM agents (continued from Algoverse).
- 2025 **SPAR Fall 2025**, *Mentor*.
3 projects, 10 mentees. Topics: goal drift in coding agents (2 papers accepted at ICLR 2026 workshops), reward hacking under unreliable tools (targeting COLM 2026), and emergent anti-sycophantic behavior.
- 2025 **Algoverse AI Safety Fellowship**, *Mentor*.
2 projects, 6 mentees: (1) how long contexts affect LLM agent safety behavior (AAAI 2026 TrustAgent workshop); (2) operationalizing situational awareness in LLM agents (ongoing via SPAR).
- 2024–25 **SPAR Spring 2025: Prompt Attacks in Unlearning**, *Mentor*.
3 mentees. Investigated robustness of unlearning methods to prompt attacks (COLM SoLaR Workshop).
- 2024–25 **AI Safety Camp 10: Multi-turn Jailbreaks**, *Research Lead*.
5 mentees. Studied the effectiveness of multi-turn jailbreak strategies (COLM SoLaR Workshop).

Talks & Community

- 2023–24 **AI Safety Fundamentals (Alignment & Governance)**, *Facilitator*, Bluedot Impact.
- 2024 **AI Safety, Ethics, and Society**, *Facilitator*, CAIS.
- 2018 **“Superguide” of the International Physics Olympiad (IPhO) 2018**, Portugal.
Responsible for 9 guides and 38 international students, and organizing activities at the IPhO.

Teaching

- 2022–23 **Techniques of Mathematical Physics**, *Physics Engineering*, IST.
- 2022–23 **Quantum Mechanics**, *Aerospace, Naval, Mechanical Engineering*, IST.
- 2017–18 **Electromagnetism and Optics**, *Aerospace, Naval Engineering*, IST.

Selected Publications

AI Safety

- 2026 **Inherited Goal Drift: Contextual Pressure Can Undermine Agentic Goals**, Menon, A. *et al* (senior author: **Cruz, D.**), ICLR 2026 AIWILD & LLA Workshops.
- 2026 **Asymmetric Goal Drift in Coding Agents Under Value Conflict**, Saebo, M. *et al* (senior author: **Cruz, D.**), ICLR 2026 AIWILD & LLA Workshops.
- 2026 **Unreliable Tools Increase Reward Hacking by AI Coding Agents**, Hadeliya, T. *et al* (senior author: **Cruz, D.**), Targeting COLM 2026.
- 2025 **When Refusals Fail: Unstable Safety Mechanisms in Long-Context LLM Agents**, (senior author: **Cruz, D.**), AAAI 2026 TrustAgent Workshop.
- 2025 **Prompt Attacks Reveal Superficial Knowledge Removal in Unlearning Methods**, Jang, Y. *et al* (senior author: **Cruz, D.**), COLM SoLaR Workshop.
- 2025 **Multi-Turn Jailbreaks Are Simpler Than They Seem**, Yang, X. *et al* (senior author: **Cruz, D.**), COLM SoLaR Workshop.
- 2024 **Understanding the Learned Look-Ahead Behavior of Chess Neural Networks**, **Cruz, D.**, TMLR.
- 2023 **Reinforcement Learning Fine-Tuning of Language Models is Biased Towards More Extractable Features**, **Cruz, D.** *et al*, NeurIPS SoLaR Workshop.

Quantum Computing

- 2019 **Efficient Quantum Algorithms for GHZ and W States, and Implementation on the IBM Quantum Computer**, **Cruz, D.** *et al*, Advanced Quantum Technologies, 1900015.
See Google Scholar for full list.

Selected Awards

- 2022 **\$3000 1st Prize in Classiq Coding Competition’s Hamiltonian Challenge**.
- 2022 **Winner of Hackathon QCHACK 2022 (QuTech Challenge)**, *Stanford*, United States.
- 2021 **Winner of Hackathon iQuHACK 2021 (D-Wave Challenge)**, *MIT*, United States.
- 2018 **2nd place for “IBM Q Best Paper Award”**, DOI: 10.1002/qute.201900015.
- 2013–14 **Astronomy Olympiad**.
1st place nationally and **Honorable Mention** at the IOAA in 2014 — **best result ever for Portugal**.
- 2013–14 **Physics Olympiad**.
Honorable Mention at the national level and at the XLV IPhO in 2014, in Kazakhstan.

Skills & Languages

Python, PyTorch, transformers, LLM evaluations infrastructure, L^AT_EX, Git.
Portuguese (native), English (fluent).